THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 2

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A person's starch digestion starts in the mouth. If saliva is absent, the digestion of: 1

(a) water stops **(b)** protein stops
(c) starch is delayed **(d)** fats stop

2. Why do adolescents have a good appetite? 1

(a) they are bored **(b)** fast growth needs more energy
(c) they are old **(d)** they are sick

3. During land breeze, warm air rises over the: 1

(a) sea **(b)** mountain **(c)** land **(d)** desert

4. A bus travels 60 km at 30 km/h. The time taken is: 1

(a) 0.5 h **(b)** 2 h **(c)** 90 h **(d)** 30 h

5. During vigorous exercise, the breathing rate: 1

(a) increases **(b)** decreases **(c)** stays same **(d)** stops

6. The pancreas is in the: 1

(a) abdomen **(b)** chest **(c)** neck **(d)** brain

7. Which of these would be best on a cold day? 1

(a) layers of woollen clothes **(b)** one thin shirt
(c) net cloth **(d)** sleeveless

8. A line in a distance-time graph passing (0,0) and (4 s, 20 m) shows speed: 1

(a) 20 m/s **(b)** 5 m/s **(c)** 80 m/s **(d)** 4 m/s

9. The heart pumps blood into the: 1

(a) arteries **(b)** food pipe **(c)** capillaries **(d)** veins

10. A child lacking iodine in his diet may develop: 1

(a) rickets **(b)** night blindness **(c)** goitre **(d)** diabetes

11. When you place your hand above a candle flame, you feel heat mainly due to: 1

(a) evaporation **(b)** conduction **(c)** radiation only **(d)** convection

12. A scooter moves at 36 km/h. Its speed in m/s is: 1

(a) 129.6 m/s **(b)** 36 m/s **(c)** 10 m/s **(d)** 3.6 m/s

13. Bile does not contain enzymes but helps in digestion by: 1

(a) breaking fat into tiny droplets **(b)** absorbing water
(c) digesting starch **(d)** digesting proteins

14. Which is the best snack for an adolescent? 1

(a) fruit and nuts **(b)** chips **(c)** cola **(d)** sweets

15. The sea is cooler than land during the day because: 1

(a) land is wet **(b)** sun does not shine on sea
(c) water takes longer to heat **(d)** water is hot

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Why are cooking pots with copper bottoms common? 2

17. Explain why speed of a cyclist uphill is less than downhill. 2

18. How do arteries differ in structure from veins? 2

19. Why is menstruation a natural process? 2

20. Why does ice keep fish fresh in a box? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. What are the effects of smoking on the respiratory system? 3

22. Explain why skin problems like pimples occur. 3

23. Describe the role of radiation in cooking on a charcoal grill. 3

24. A student measured the time taken by a ball to roll 2 m as 4 s and then 6 m as 8 s. Compare the speeds. 3

OR

What are the digestive problems caused by poor eating habits? 3

25. Explain the sequence of events in a menstrual cycle in short. 3

OR

Explain why sea breeze is stronger in the afternoon. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain the emotional changes in adolescence and give guidance to teenagers for managing them. 5

OR

Describe the construction of a solar water heater and explain the heat transfer in it. 5

27. Explain why a pendulum clock stops after a while and how it is kept going. 5

OR

Explain how the respiratory organs differ in humans, fish, insects, earthworms and amoeba. 5

END OF QUESTION PAPER — Total: 50 Marks

MINDSTEP MIND